

Induction for *Nat*

The set *Nat* can be defined by the following grammar:

$$N \in \mathit{Nat} ::= \mathit{zero} \mid \mathit{succ}(N).$$

This gives rise to the following inductive principle, for any $P \subseteq \mathbb{N}$:

$$P(\mathit{zero})$$

$$\wedge$$

$$\forall n \in \mathit{Nat}. [P(n) \Rightarrow P(\mathit{succ}(n))]$$

$$\implies$$

$$\forall n \in \mathit{Nat}. [P(n)]$$